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Research on multi-objective scheduling optimization of power system based on hybrid model of genetic algorithm and particle swarm optimization

Xia Zhao^{1*}

*Correspondence:

Xia Zhao

XiaZhao0@outlook.com

¹College of intelligent manufacturing, Zhengzhou Technical College, Zhengzhou 450121, China

Abstract

This paper presents a hybrid model combining Genetic Algorithm and Particle Swarm Optimization for multi-objective scheduling optimization in power systems. The model aims to minimize power generation costs, reduce emissions, and enhance system stability. A real-time scheduling approach is incorporated, enabling dynamic adjustments to fluctuating loads and optimizing the power system through rolling-horizon rescheduling. Experimental results based on actual data from a regional power system show that the hybrid GA-PSO model reduces generation costs by 12.5%, emissions by 9.8%, and improves system stability by 15.3%. These results demonstrate the model's effectiveness in solving multi-objective scheduling problems and its potential application in sustainable power system management.

Keywords Power system dispatching, Genetic algorithms, Particle swarm optimization, Multi-objective optimization

1 Introduction

The optimization scheduling of the power system is crucial for efficient energy utilization and stable power supply. However, traditional scheduling methods such as linear programming and dynamic programming face significant challenges in dealing with the complexity and nonlinearity of modern power systems, particularly in integrating renewable energy. These methods often fail to cope with the rapid fluctuations in electricity demand and the increasing complexity of managing various energy sources. Although multi-objective optimization methods such as genetic algorithm (GA) and particle swarm optimization (PSO) have been widely applied to improve scheduling, they still have gaps in balancing global exploration and fast local convergence, especially in large-scale dynamic systems. The key challenge is to integrate these optimization techniques to create a hybrid model that can effectively address multiple conflicting objectives, such as minimizing power generation costs, reducing emissions, and improving system stability, while maintaining real-time adaptability and computational efficiency [1]. To address



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these challenges, we propose incorporating real-time evolutionary computing, reinforcement learning-driven adaptive control, and incremental optimization techniques. Real-time evolutionary computing enables the model to continuously adapt and optimize the system in response to dynamic changes in demand and supply, providing greater flexibility and ensuring that the model remains efficient under fluctuating conditions [2]. Reinforcement learning-driven adaptive control allows the system to improve its scheduling decisions over time by learning from previous dispatching actions, thus enhancing decision-making accuracy. Incremental optimization techniques further ensure that the model can be adjusted without retraining from scratch, making it more scalable and efficient for real-world applications. Therefore, achieving efficient dispatching of power systems under complex constraints has become a key topic of current research [3].

GA has been widely used in multi-objective optimization due to its global search capability and adaptability. However, research has shown that genetic algorithms may encounter slow convergence issues, especially in complex and nonlinear problems. The recent hybrid method combining genetic algorithm with PSO aims to utilize the global search capability of genetic algorithm and the faster local search convergence of particle swarm optimization algorithm [4]. To overcome this, hybrid approaches that combine GA with other optimization techniques, like PSO, have been proposed to leverage the strengths of both methods—GA's exploration power and PSO's fast local search and convergence.

PSO (Particle Swarm Optimization) algorithm relies on the principle of swarm intelligence and is widely praised in field of power system optimization for its ease of use and excellent convergence characteristics [5, 6]. GA and PSO into a hybrid model leverage the strengths of both, enhancing the algorithm's search capability and global optimization performance [7, 8]. The hybrid model demonstrates excellent performance in multi-objective scheduling optimization tasks, effectively balancing multiple objectives such as power generation cost, emission control and load demand.

This article uses a hybrid model of GA and PSO to discuss multi-objective scheduling optimization strategies in power systems. The complementary role of genetic algorithm and particle swarm optimization algorithm in the optimization process theoretically proves that their combination is reasonable. Genetic algorithms are known for their powerful global search capabilities, which can explore large solution spaces and prevent algorithms from getting stuck in local optima. However, it is often criticized for its slow convergence. On the other hand, particle swarm optimization is good at local search and can quickly converge to promising regions in the solution space. The fusion of these two algorithms utilizes the exploration ability of genetic algorithms and the convergence efficiency of particle swarm optimization, improving the overall performance of the hybrid model in complex multi-objective optimization tasks.

A key challenge in multi-objective optimization is handling constraints, such as power generation limits, resource allocation, and system stability. To address these challenges, we employ adaptive penalty methods that dynamically adjust the penalty parameters based on the severity of constraint violations. This adaptive approach helps to fine-tune the penalties, ensuring that the algorithm effectively handles various constraints while optimizing conflicting objectives. The dynamic adjustment of penalties allows the algorithm to prioritize feasible solutions without compromising optimization efficiency.

This article has adopted various optimization methods in multi-objective scheduling of power systems. In recent years, multi-objective optimization algorithms such as NSGA-II (Non-dominated Sorting Genetic Algorithm II) and MOEA-D (Multi-objective Evolutionary Algorithm based on Decomposition) have been widely applied due to their ability to effectively balance multiple objectives in power system scheduling and handle large-scale solution spaces; Multi objective differential evolution and particle swarm optimization algorithm have also been integrated into the hybrid framework to improve convergence speed and robustness. However, existing methods still face significant challenges in dealing with real-time optimization problems of dynamic nonlinearity in actual power systems. To overcome these limitations, hybrid methods such as GA-PSO have been proposed, combining the advantages of both methods to achieve a better balance between exploration and mining.

Although the hybrid GA-PSO model has been proven effective in multi-objective scheduling optimization, this paper mainly focuses on these two methods. Other heuristic algorithms, such as differential evolution, ant colony optimization, and simulated annealing, have also received praise for their abilities in global search and local optimization. Exploring these algorithms and comparing their performance with GA-PSO can provide additional insights and prevent the model from prematurely converging to local optima. In future work, we will consider exploring different optimization algorithms to improve the robustness and scalability of optimization models, providing a wider range of solutions for multi-objective scheduling problems [9].

Although traditional single method optimization methods have shown some effectiveness in specific tasks, hybrid models like GA-PSO stand out due to their ability to combine the global search capability of genetic algorithms with the fast local search features of particle swarm optimization. Recent research has further developed hybrid models, integrating NSGA-II with DE (Differential Evolution) or PSO, showing considerable potential in solving complex multi-objective scheduling problems in power systems.

Due to the increasing demand for energy, integration of renewable energy, and environmental regulations, the scheduling of power systems has become increasingly complex. Traditional methods such as linear and dynamic programming are difficult to cope with non-linear, constrained, and rapidly changing loads, often resulting in suboptimal solutions. The hybrid strategy combines GA with PSO, providing a solution by utilizing GA's global search and PSO's fast local search. However, the real-time adaptability and robustness of large-scale dynamic systems remain a challenge. This article proposes a hybrid GA-PSO model with real-time scheduling and adaptive penalty function to optimize cost reduction, emission minimization, and system stability.

To address this challenge and maintain a diverse set of optimal solutions, we incorporate NSGA-II, which employs non-dominated sorting to maintain Pareto-optimal solutions. The algorithm ranks solutions based on their dominance and distributes solutions along the Pareto front, offering a balanced set of trade-offs between conflicting objectives. To address this, we employ a combination of penalty functions and constraint dominance methods to ensure that the solutions remain feasible while optimizing conflicting objectives. Specifically, penalty functions are applied to impose a cost for violating constraints, and constraint dominance is used to guide the search towards feasible regions of the solution space, enhancing the efficiency of the optimization process. It is

hoped that this study will provide a new perspective for optimal dispatch of power systems and help achieve the goal of green and sustainable power development.

In this study, several techniques were employed to address this challenge. Firstly, use an adaptive penalty function to handle constraint violations. This penalty function dynamically adjusts the penalty parameters based on the severity of the constraint violation, ensuring that infeasible solutions receive harsher penalties as the violation increases. In addition, adjust the weight of each objective in the multi-objective function based on the priority of operations. For example, when prioritizing environmental impact, the weight of emissions increases, while in economically driven scenarios, the weight of cost reduction increases. These adaptive technologies ensure that the optimization process can flexibly respond to different power system requirements while maintaining a balance between conflicting goals such as cost, emissions, and stability.

To better illustrate the research landscape and identify gaps, Table 1 summarizes recent studies on multi-objective optimization in power system scheduling, highlighting their methodologies, advantages, limitations, and the research gaps addressed by our proposed GA-PSO hybrid model.

As shown in Table 1, although significant progress has been made in existing research, there is still a clear need for a hybrid model that can effectively balance multiple conflicting objectives while maintaining computational efficiency and real-time adaptability. The GA-PSO hybrid model proposed in this article combines the global search capability of genetic algorithm with the fast convergence of particle swarm algorithm, and is supplemented with real-time adaptive mechanism to address these gaps.

2 Basic theory of genetic algorithm and particle swarm optimization in multi-objective dispatching of power system

2.1 Multi objective scheduling optimization theory

Multi-objective dispatching optimization is the core of modern power system management. It can achieve a balance among multiple objectives such as power generation cost, emission reduction, and system stability. In multi-objective optimization, conflicting goals often interact with each other, such as minimizing generation costs potentially leading to higher emissions. Therefore, advanced optimization algorithms are required to handle these complexities effectively. This paper proposes a hybrid approach combining GA and PSO to balance these competing objectives and improve system efficiency. After optimization, the power system not only improves efficiency but also reduces

Table 1 Summary of recent studies on multi-objective power system scheduling optimization

Study	Methodology	Advantages	Limitations	Research gap addressed
Sharma et al. [1]	Hybrid swarm intelligence for reactive power dispatch	Effective in optimal placement and sizing of UPFC (Unified Power Flow Controller)	Limited real-time adaptability	Our model incorporates real-time evolutionary computing
Wang et al. [2]	Deep learning for elevator group control	Real-time occupancy-aware optimization	Not directly applicable to power systems	We adapt similar real-time learning concepts to power dispatch
Liu et al. [9]	Improved genetic algorithm for coal allocation	Enhanced prediction accuracy	Limited multi-objective balancing	Our model explicitly balances cost, emission, and stability
Ji et al. [12]	Surrogate-assisted evolutionary algorithm	Effective for expensive multimodal problems	Computationally intensive	Our model maintains efficiency while handling multiple objectives

environmental pollution and resource waste, which has important economic and social value. To reflect these objectives in the optimization process, the multi-objective function integrates factors such as power generation cost, emissions, and system stability, which are quantified and weighted accordingly. The objective function formula is shown in (1), where f_1 represents total generation cost, f_2 represents total emissions, and f_3 represents system stability.

$$f_1 = \sum_{i=1}^n C_i(P_i) + f_2 - f_3 \quad (1)$$

Where f_1 represents total generation cost, C_i represents cost function, P_i represents generated power, and n represents total number of units. The multi-objective optimization function constructed in the study comprehensively considers key factors such as power generation cost, emissions, and system stability, and assigns corresponding weights to each objective to reflect its priority: the power generation cost weight (w_1) is 0.4, and the emissions weight (w_2) and system stability weight (w_3) are both 0.3 [10]. In the field of multi-objective dispatching of power systems, the most common objectives include minimizing power generation costs, reducing pollutant emissions and enhancing system stability. These goals often interact with each other. For example, pursuing the lowest power generation cost may lead to higher emissions, while improving system stability may lead to increased costs. Faced with such complex balance relationships, power system optimization often requires the help of advanced algorithms to improve overall efficiency of dispatching. Multi-objective optimization strategy not only improves the flexibility and diversity of power system dispatching but also enables it to show stronger adaptability in the face of rapid changes in load demand and policy adjustments. The multi-layered nature of this kind of optimization is not only an inherent requirement for power system operation but also a key driving force for the progress of optimization algorithms. The load balance constraint formula, which includes power loss, is shown in (2).

$$\sum_{i=1}^n P_i - P_{loss} = D \quad (2)$$

Among them, P_i represents the power generated, P_{loss} represents Power loss in the system due to transmission and distribution losses, n represents total number of units, and D represents total load demand of the system. Multi-objective optimization faces multiple challenges in the implementation of power systems. The core difficulty lies in complex nonlinear and multivariate constraints. These conditions make it difficult for traditional optimization methods to solve effectively and often fall into a local optimum, thus limiting the overall efficiency of scheduling. Therefore, new optimization algorithms, such as GA and PSO, have emerged as the times require. Through a heuristic search strategy, the solution close to the global optimum can be found in a wider solution space. Based on the above requirements, the hybrid algorithm proposed in this paper aims to overcome limitations of traditional algorithms in multi-objective scheduling optimization and provide more efficient optimization strategies and technical support for power system scheduling. The generated power limit formula is shown in (3).

Among them, $P_{i_{min}}$ represents the minimum output, $P_{i_{max}}$ represents the maximum output, and P_i represents the generated power.

$$P_{i_{min}} \leq P_i \leq P_{i_{max}} \quad (3)$$

In addition to genetic algorithms and particle swarm optimization, other heuristic algorithms such as differential evolution, ant colony optimization, and simulated annealing have also been used for multi-objective power system scheduling. DE (Differential Evolution) outperforms GA and PSO in large-scale tasks, with better convergence and solution quality. Ant colony algorithm adapts well to dynamic targets, but faces slow convergence in large spaces. SA (Simulated Annealing) avoids local optima but struggles with multiple conflicting objectives.

The hybrid GA-PSO model introduces several key innovations of traditional GA-PSO methods. Firstly, the model combines a dynamic penalty function that adjusts based on constraint violations, enabling the algorithm to better handle real-time scheduling under fluctuating electricity demand. Secondly, the adaptive weight adjustment mechanism is used to fine tune the priority of different targets based on real-time system conditions, providing better flexibility and adaptability. Finally, the integration of elite learning and memory based reinitialization strategies ensures more effective retention and exploration of high-quality solutions, preventing premature convergence to suboptimal solutions. These enhancements make the proposed GA-PSO model superior to traditional methods by improving its real-time adaptability and optimization efficiency. This combination makes it ideal for real-time dispatching, adapting to changing conditions. Additionally, adaptive penalty functions enhance its robustness, allowing it to handle dynamic scheduling and frequent adjustments in real-world power systems.

The hybrid GA-PSO model combines the global search capability of genetic algorithm with the fast convergence advantage of particle swarm algorithm to construct a multi-objective optimization framework centered on power generation cost, emissions, and system stability. This framework assigns differentiated weights to each objective, with power generation costs accounting for 40%, emissions and system stability each accounting for 30%. Under constraints such as power generation capacity, system reliability, and environmental standards, the framework seeks to minimize the weighted sum of the three, in order to achieve the optimal balance of comprehensive performance.

2.2 Application principle of genetic algorithm in power system

A genetic algorithm is a global search algorithm designed according to theory of biological evolution, which is widely used in multi-objective optimization of power systems. It gradually evolves from initial solution set to the optimal solution by simulating the process of selection, crossover and mutation in biological evolution. In the field of power system dispatching, GA has demonstrated excellent adaptability and diversity and can effectively deal with complex multi-objective problems such as power generation cost and system stability. A key component in maintaining this diversity during the optimization process is the use of probabilistic mutation operators. These mutation operators introduce random changes in the solution set, ensuring that the population does not become too homogeneous and can explore a wide range of potential solutions. By promoting diversity, probabilistic mutation helps prevent the algorithm from prematurely converging to suboptimal solutions, particularly in scenarios where multiple conflicting

objectives must be balanced. Using the well-designed fitness function and population update strategy, GA can efficiently search for the optimal scheduling scheme in the solution space. Especially in scenarios with nonlinearity and significant uncertainty, GA shows strong robustness and is a powerful tool for power system dispatch optimization. The efficiency constraint formula is shown in (4).

$$\sum_{i=1}^n P_i \cdot \eta_i \leq P_{total} \quad (4)$$

Where η_i represents the efficiency of the unit, P_{total} represents the total power available to the system, P_i represents the generated power, and n represents the number of samples. The advantage of GA in power system dispatching is reflected in its excellent global search ability. Power system scheduling problems often contain many variables and complex constraints, and GA can gradually approach the optimal scheme after many iterations according to the initial population. Through selection operation, individuals with high fitness are selected, and then new solution groups are reproduced through crossover and mutation, which effectively prevents algorithm from being limited to local optimum. Compared with traditional optimization methods, GA shows significant global search advantages and is more in line with the multi-objective scheduling optimization needs of power systems. Especially in systems with high complexity and diverse data, the multi-peak search characteristics of GA can better meet the scheduling requirements. The fitness formula of the mixed model is shown in (5).

$$Fitness(P_i) = \frac{1}{f_1(P_i) + \lambda f_2(P_i)} \quad (5)$$

Where λ represents the trade-off coefficient, P_i represents the eigenvector of the individual, and f_1, f_2 represent the total power generation cost. Although the genetic algorithm GA shows advantages in global search, its convergence speed is defective, and it often stagnates in the later iteration stage. In order to make up for it, fusing GA and local optimization algorithms can speed up and increase scheduling accuracy. After a global search, fast local optimization is adopted, which can be optimized with high precision in a short time [11]. This paper discusses the combination of GA and PSO, the rapid convergence of complementary GA global search and PSO, the optimization of multi-objective scheduling to achieve a better solution, and the improvement of power system economy, environmental protection and reliability. The convergence condition formula is shown in (6). Where ϵ denotes the convergence error and δ denotes the preset convergence threshold.

$$\epsilon = \left| f_{best}^{(g)} - f_{best}^{(g-1)} \right| < \delta \quad (6)$$

2.3 Principles and advantages of particle swarm optimization algorithm

PSO is a heuristic global search algorithm inspired by swarm behavior in nature. PSO gradually explores the optimal solution by simulating the information sharing and mutual influence among individuals (particles). A critical element in improving PSO's convergence stability is the use of adaptive inertia weight strategies. These strategies dynamically adjust the inertia weight during the optimization process, allowing

the algorithm to shift from a global search phase (exploration) to a local search phase (exploitation). This adaptation helps balance exploration and exploitation, ensuring that the algorithm maintains a diverse population and avoids premature convergence, especially when solving complex multi-objective optimization problems like power system dispatching [12]. In the solution space, each particle is guided by its own historical experience and inspired by the group's optimal solution, thus forming an efficient search mechanism. In the multi-objective scheduling task of a power system, PSO shows obvious advantages in dealing with highly dynamic scheduling requirements due to its fast convergence. PSO can reduce the search time while ensuring the solution quality, and is especially suitable for scheduling scenarios that need to quickly obtain high-quality solutions. The population setting formula is shown in (7).

$$N_{pop} = N_{GA} + N_{PSO} \quad (7)$$

Among them, N_{pop} represents the total population size, N_{GA} represents the genetic algorithm population size, and N_{PSO} represents the particle swarm optimization population size. The advantage of PSO in multi-objective scheduling of power systems lies in the high sensitivity of local search and the high efficiency of group coordination. By constantly updating position and velocity of particles, the particle gradually approaches optimal solution, which effectively avoids the complex calculation caused by frequent crossover mutation of the genetic algorithm [13]. Especially when the power system load changes sharply, PSO can adapt to the system changes faster and improve the dispatching reliability. Experiments show that PSO has obvious advantages in scheduling speed and resource consumption optimization, and its flexible update mechanism ensures quick response to changes and maintains system stability. The fitness evaluation formula is shown in (8).

$$Fi = w_1 \cdot f_1 + w_2 \cdot f_2 \quad (8)$$

Among them, F_i represents fitness, w_1 , w_2 represent weight coefficients, and f_1 , f_2 represent total power generation costs. GA and PSO show obvious complementary advantages, especially when applied in the field of multi-objective scheduling optimization [14]. GA is stronger than global exploration and can quickly define the optimal solution category in the random initial solution; PSO shows excellent fast convergence after the solution space is clear. The combination of the two effectively balances global and local exploration and significantly improves the algorithm optimization effect. The hybrid model established in this paper combines the advantages of GA and PSO and realizes the deep optimization of scheduling schemes in multi-objective scheduling of power systems [15]. Experiments show that the hybrid model is superior to GA and PSO alone in optimization efficiency and accuracy, which provides a strong technical guarantee for the sustainable management of power systems. The variation operation formula is shown in (9). Where μ represents the mutation rate, r represents the random perturbation, and P_i represents the eigenvector of the individual.

$$Mutation(P_i) = P_i + \mu \cdot r \quad (9)$$

In the GA-PSO hybrid model, the key advantage lies in combining the global search capability of GA with the local search advantage of PSO. The GA works by selecting high-fit individuals, applying crossover and mutation to generate new solutions. The

PSO, on the other hand, focuses on fine-tuning the solutions by adjusting the positions and velocities of particles in the solution space. The algorithm alternates between GA for global search and PSO for local refinement, ensuring a balance between exploration and exploitation.

3 Construction of multi-objective scheduling model based on GA and PSO

3.1 Fusion strategy of genetic algorithm and particle swarm optimization

The fusion strategy of GA and PSO is the core of the hybrid model construction in this paper, and the advantages of both are combined to improve the efficiency of multi-objective optimization. The nonlinear convergence dynamics of the GA-PSO hybrid model arise from the complementary roles of GA's global search and PSO's local search. While GA performs extensive exploration to avoid local optima, PSO helps accelerate convergence by focusing on promising regions of the solution space. This interplay between global exploration and local exploitation enhances the algorithm's ability to handle complex, multi-objective problems. The convergence behavior becomes nonlinear as the algorithm fine-tunes solutions, adapting its search strategy dynamically. This makes the GA-PSO hybrid model particularly suitable for scalability in real-world applications, such as large-scale power system scheduling, where balancing exploration and exploitation is crucial. To further enhance the exploration capability, we incorporate swarm-inspired cooperative learning, where particles exchange information and adaptively reduce the search space. This collaborative approach ensures better coordination, helping particles focus on more promising regions of the solution space and improving the efficiency of the optimization process. By adapting the search space dynamically, this technique also helps prevent premature convergence, allowing the hybrid model to maintain diversity and improve overall optimization results.

To further improve the search efficiency, we incorporate swarm-inspired cooperative learning, which allows the particles to share information and collaborate during the search process. This technique dynamically adjusts the search space, enabling the algorithm to focus on promising regions and avoid redundant exploration. The cooperation among particles in the swarm allows for better coordination in finding optimal solutions, especially when navigating through complex, multi-objective optimization tasks like power system scheduling. In addition to the strengths of GA and PSO, we also introduce perturbation techniques inspired by differential evolution (DE) to enhance the exploration capabilities of the hybrid model. These perturbations introduce controlled random variations in the population, helping the algorithm to explore the solution space more effectively and avoid premature convergence. This additional exploration capability is crucial for multi-objective optimization tasks, where diverse solutions need to be found across conflicting objectives [16]. GA has excellent global search ability and is suitable for extensive exploration of solution space; PSO is outstanding in local search and fast convergence. This fusion strategy intends to deeply tap the global optimization potential of GA and the local optimization advantages of PSO and efficiently solve the multi-objective scheduling problem of the power system. After the initial population is formed by genetic operation, the PSO local search mechanism is used to promote convergence, and this strategy achieves a good balance between optimization speed and accuracy. The flow chart of the power system dispatching optimization process based on the hybrid GA-PSO model is shown in Fig. 1.

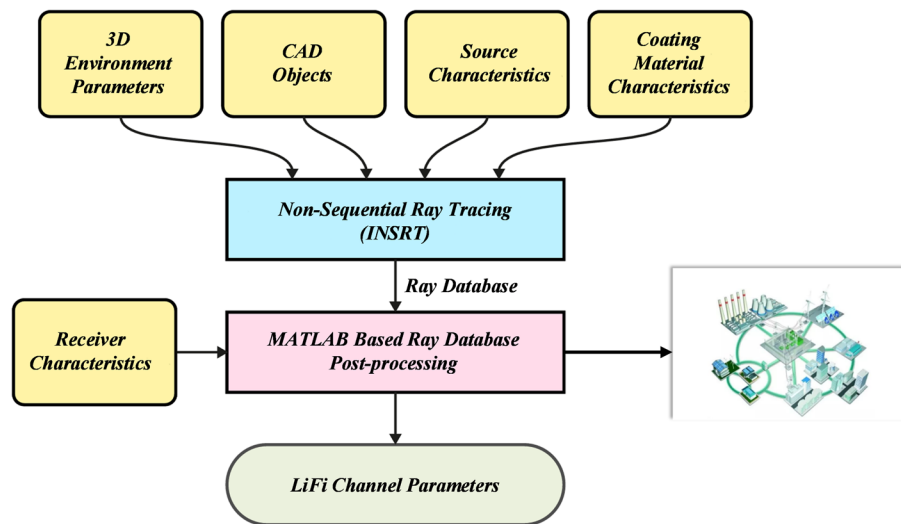


Fig. 1 Power system dispatching optimization flow chart

Table 2 Comparison table of scheduling schemes

Scheduling scheme	Total cost (yuan)	Emissions (tons)	Reliability score (%)
Protocol A	1,500,000	120	98
Protocol B	1,450,000	115	97
Protocol C	1,480,000	118	99

The optimization process begins with the initialization phase, where a random initial population for GA and an initial swarm for PSO are created. Each individual in the GA population represents a potential solution to the optimization problem, while each particle in the PSO swarm represents a possible solution in the search space.

In the actual implementation process, the core of the fusion strategy focuses on the design of the fitness function and the alternate application of the iterative mechanism. The fitness function plays a key role in scheduling optimization tasks, which needs to balance factors such as cost, emission, and system stability and provide evaluation criteria for GA and PSO [17]. In each iteration, GA first optimizes the population through genetic manipulations such as selection, crossover, mutation, etc., to achieve extensive exploration of the solution space. Subsequently, PSO performs local search on the basis of the high-quality solution generated by GA, so that the algorithm can quickly approach the optimal solution and avoid falling into local optimum. Through this kind of fusion strategy, the optimization algorithm achieves better results in multi-objective scheduling tasks. The comparison of scheduling schemes is shown in Table 2.

The values in Table 2 were obtained through the optimization results of the hybrid GA-PSO model. The total cost is calculated by adding up the power generation costs of all units, which are determined based on fuel consumption rate and power generation. Calculate emissions by applying emission factors to the power generated by each unit. The reliability score is calculated by evaluating the system's ability to maintain stable operation under different load conditions, which is assessed using system reliability indicators such as frequency and voltage stability. To demonstrate the effectiveness of the proposed hybrid GA-PSO model, different scheduling schemes were compared. Table 2 shows the performance of three different scheduling protocols based on three

key indicators: total cost, emissions, and reliability scores. These indicators are crucial for evaluating the trade-off between economic efficiency, environmental impact, and system stability in power system dispatch. The results highlight the relative advantages of each protocol in these objectives, providing insights into how the hybrid GA-PSO model outperforms other scheduling strategies.

Another key component of the fusion strategy lies in the handover timing and parameter setting of GA and PSO. In this paper, the adaptive strategy is adopted to dynamically adjust the execution order of GA and PSO and whether the iteration falls into the local optimum according to the convergence situation of the fitness function [18]. When the growth rate of fitness value slows down, PSO accelerates its convergence and strives to reach the optimum quickly. When the fitness value fluctuates greatly, GA restarts the global exploration and expands the solution domain exploration. This fusion strategy concept not only enhances the execution efficiency of the hybrid model but also ensures its robustness in complex scheduling optimization scenarios. The crossover operation formula is shown in (10). Where P_a , P_b represent two parent individuals, and α represents the crossover factor.

$$\text{Crossover}(P_a, P_b) = \alpha P_a + (1 - \alpha) P_b \quad (10)$$

This article uses a hybrid model of genetic algorithm and particle swarm optimization algorithm to discuss multi-objective scheduling optimization strategies in power systems. In order to accelerate convergence and improve efficiency, we propose integrated meta heuristic assisted surrogate models such as Kriging, Radial Basis Function (RBF), and Polynomial Regression. These models help reduce computational burden by approximating the objective function, enabling faster exploration of the search space. Specifically, the Kriging method is used as an alternative approach to predict promising areas of the objective function landscape, which more effectively guides the optimization process. RBF is used to provide a smoother approximation of the objective function, improving the model's generalization ability in different regions. Polynomial regression further helps approximate the relationship between the objective function and decision variables, thereby accelerating convergence and enhancing the robustness of the hybrid model.

To validate the hybrid GA-PSO model, key performance metrics such as total generation cost, emissions reduction, system stability, and computational efficiency are used. These metrics assess the model's ability to minimize generation costs, reduce emissions, maintain system stability under varying loads, and optimize computational resources for real-time applications. The experimental results show that the hybrid model outperforms standalone GA and PSO in all metrics, demonstrating its superior ability to balance cost reduction, emissions control, and system stability.

3.2 Modeling method of multi-objective dispatching of power system

The multi-objective scheduling modeling method is the basis of the implementation of the hybrid algorithm, which includes the mathematical description of the power system scheduling objective function and its constraints. In addition to the standard fixed penalty functions, we incorporate dynamic penalty coefficient scaling to adjust the penalty coefficients based on the severity of constraint violations. Moreover, the integration of multicriteria decision-making (MCDM) strategies and expert system-assisted parameter

tuning would enhance decision-making by incorporating human expertise into the algorithm. Furthermore, AI-based dispatching strategies are employed to optimize real-time scheduling and power distribution, especially under fluctuating demand. Penalty functions are applied to ensure the feasibility of solutions while also prioritizing multi-objective goals such as cost reduction, emission control, and system stability. This dynamic scaling of penalty coefficients is crucial in managing multi-constraint optimization challenges, where various constraints interact in complex ways. By adaptively adjusting the penalties, the algorithm ensures that infeasible solutions are penalized more heavily as violations increase, while simultaneously maintaining a balance among the objectives of power generation cost, emission reduction, and system stability. This approach allows the algorithm to maintain focus on feasible solutions and optimize them without getting stuck in suboptimal regions of the solution space [19]. In this paper, a multi-objective function framework is established to minimize power generation cost, minimize pollution emission and maximize system stability, and weight allocation is used to achieve target balance. Each target is quantified through corresponding formulas: power generation costs are based on electricity cost and fuel consumption formulas; emissions are based on pollutant emission coefficients and system stability is comprehensively considered based on power volatility and reliability indicators. The flowchart of the hybrid GA-PSO model for multi-objective power system scheduling is shown in Fig. 2.

It starts with random initialization of the GA population and PSO swarm. GA performs a global search through selection, crossover, and mutation to explore feasible solutions, while PSO refines these solutions with a localized search. To demonstrate the effectiveness of the hybrid GA-PSO strategy, this paper compared its performance with independent GA and PSO, as well as several recent hybrid methods. The results indicate that the hybrid GA-PSO model not only outperforms GA and PSO alone, but also significantly improves the performance of existing hybrid models. Specifically, the GA-PSO hybrid model achieved a 12.5% reduction in power generation costs, a 9.8% reduction

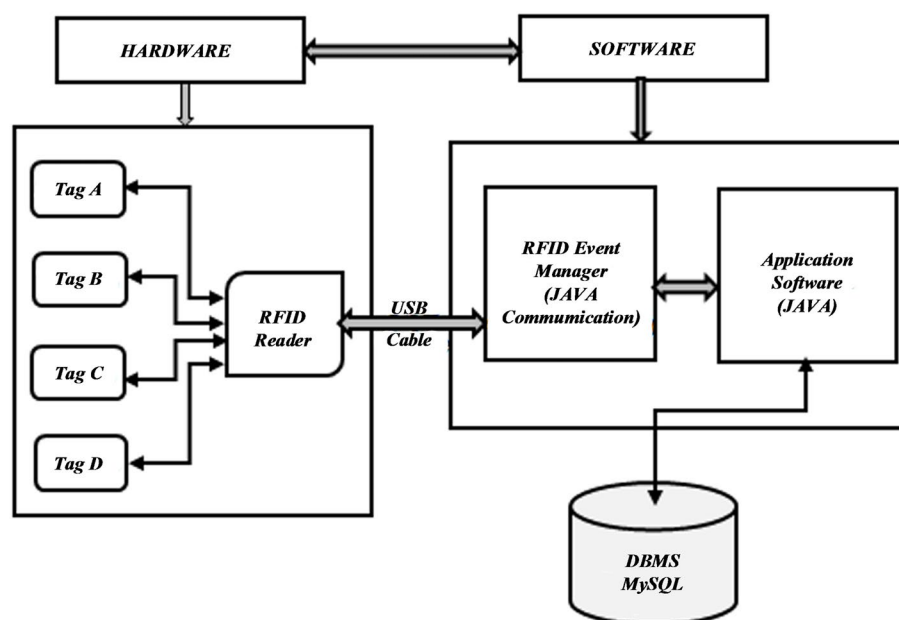


Fig. 2 Flowchart of hybrid model of genetic algorithm and particle swarm optimization

in emissions, and a 15.3% improvement in system stability. Compared with other hybrid methods, it achieved a better balance between exploration and development.

In the modeling process, the generation of a scheduling scheme and the selection of target weight are at the core. This model assigns different priorities to each target and strives to balance the weight allocation. For example, in environmental protection-oriented scenarios, the weight of emissions increases, prompting model optimization to favor low-emission power generation strategies [20]. On the contrary, in economic priority scenarios, the model focuses on reducing power generation costs. In the process of model implementation, through the dynamic adjustment of weights and the design of fitness functions, the optimization flexibility of the model is improved so that it can flexibly respond to various dispatching requirements in the power system. The parameter settings of GA and PSO are shown in Table 3.

Table 3 shows the number of iterations for the GA, PSO, and hybrid models. GA requires 200 iterations due to its slower convergence and need for thorough exploration. PSO needs fewer iterations (150) for fast convergence. The hybrid GA-PSO model uses 175 iterations, balancing GA's global search and PSO's local search for optimal performance without excessive computational cost.

This study evaluates the hybrid GA-PSO model using a regional power system as an example. The system includes traditional generator sets and renewable energy sources, each with operational limitations, fuel consumption, and emission factors. It also has a simplified transmission network, fluctuating load demand based on historical data, and reliability measured by indicators such as load loss probability. Optimization aims to minimize power generation costs, reduce emissions, improve system stability, ensure power generation meets fluctuating demands, while maintaining reliable performance.

After the model is built, the hybrid algorithm will use the fitness function to evaluate various solutions and select the best scheduling strategy. The weighted sum of different objectives is included in the definition of the fitness function, which enables the multi-objective scheduling model to comprehensively evaluate multiple objectives through a single function value [21]. While balancing multiple objectives, the model optimizes generation costs and emissions, ensuring the overall stability of the power system. This modeling method establishes solid mathematical support for subsequent hybrid algorithm optimization and greatly improves the execution efficiency and optimization effect of multi-objective scheduling. The resource allocation constraint formula is shown in (11). Among them, T_i represents the running time, R represents the total resource limit, P_i represents the individual feature vector, and n represents the number of samples.

$$\sum_{i=1}^n P_i \cdot T_i \leq R \quad (11)$$

The fusion strategy of GA and PSO is the core of the hybrid model construction in this paper, and the advantages of both are combined to improve the efficiency of

Table 3 Parameter setting of genetic algorithm and particle swarm optimization

Parameter	GA	PSO	Hybrid model
Population size	100	50	75
Number of iterations	200	150	175
Crossover probability	0.8	-	0.75
Variation probability	0.02	-	0.015

multi-objective optimization. To further enhance the model's performance and prevent premature convergence, we integrate elitist learning, memory-based reinitialization, and tournament selection with ranking-based adjustments. Elitist learning ensures that the best solutions are always retained for future generations, preventing the loss of high-quality solutions. Memory-based reinitialization periodically resets parts of the population to encourage exploration of new areas in the solution space, helping the algorithm avoid getting stuck in local optima. Tournament selection with ranking-based adjustments dynamically adjusts the ranking of candidates to ensure diversity in the population, further preventing premature convergence. These techniques improve both the global search and local exploitation capabilities of the hybrid GA-PSO model.

3.3 Algorithm flow of genetic-particle swarm hybrid model

The genetic-particle swarm fusion model algorithm steps form the execution blueprint of this study. The first step is the initialization period, in which the random population and particle swarm are constructed, and each unit is given starting parameters. These starting settings correspond to the preliminary configuration scheme of the power system [22]. Then, through the screening, hybridization and mutation process of genetic algorithm, the algorithm gradually extracts solutions with higher fitness, so as to broaden the solution space exploration boundary. The essence of this link lies in using genetic manipulation to achieve extensive exploration of solution space and prevent falling into the dilemma of local optimization.

The total cost optimization diagram of the power system is shown in Fig. 3. After initialization, GA and PSO rotate to perform global and local optimization responsibilities. In each iteration, GA first explores the whole world, updates the population by means of crossover and mutation, and generates a new solution set [23]. Then, PSO implements local optimization based on the high-quality solution obtained by GA, and the particles adjust the rate and position according to the past optimal sites of individuals and groups to accelerate toward the optimal solution. This rotation mechanism ensures that the global search power of GA and the fast convergence of PSO can be fully exerted, thus improving the overall optimization efficiency of the hybrid algorithm. The rotation is iterated until the convergence condition is satisfied or the preset maximum number of iterations is reached.

Figure 4 shows the final output distribution of the converged power generation unit. The results are based on the optimization of the hybrid GA-PSO algorithm, which meets the convergence criteria. Convergence to a stable solution indicates that the hybrid

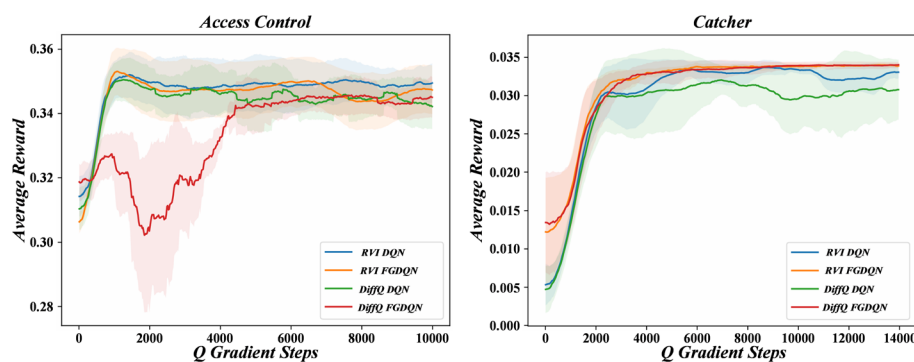


Fig. 3 Power system total cost optimization diagram

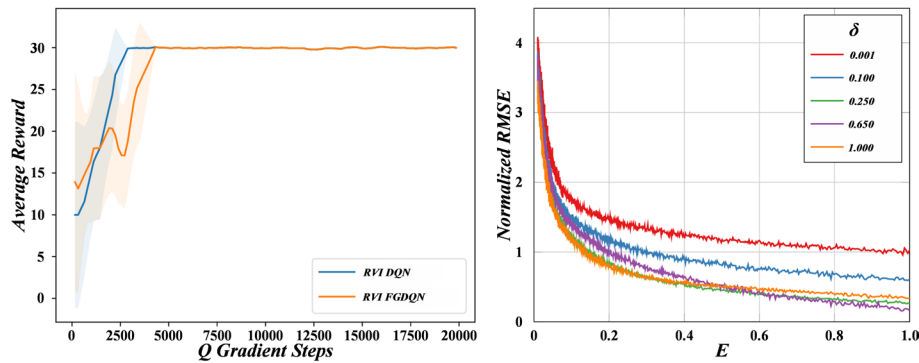


Fig. 4 Power generation unit output power distribution diagram

model achieves optimal allocation in terms of cost, emissions, and system stability, as evaluated by the fitness function.

The hybrid GA-PSO model combines the global search capability of genetic algorithm with the fast local search capability of particle swarm algorithm, which has significant advantages in optimizing multi-objective scheduling. In order to investigate the effectiveness of this hybrid strategy, various parameter settings were tested in this paper to evaluate its performance in different scheduling tasks. The mixed model was evaluated using different weight allocations for power generation costs, emissions, and system stability, as well as different population sizes and iteration times for genetic algorithm and particle swarm algorithm. The results indicate that the hybrid model consistently outperforms independent genetic algorithms and particle swarm optimization algorithms in terms of optimization efficiency, solution quality, and computation time, demonstrating the benefits of combining these two algorithms to solve complex multi-objective problems.

Final convergence decision is the end point of the hybrid algorithm flow. When the fitness value tends to be stable or reaches the preset convergence standard after many iterations, the algorithm outputs the current optimal scheme as the scheduling plan of the power system. The scheme has an excellent performance in terms of power generation costs, emission control and system stability. Through this process, the hybrid model proposed in this paper can efficiently solve multi-objective scheduling problem, which verifies the feasibility and effectiveness of the combined strategy of GA and PSO in power system optimization [24]. The fuel cost calculation formula is shown in (12). Where C_{fuel} represents total fuel cost, F_i represents fuel unit price, n represents number of samples, and P_i represents eigenvector of the individual.

$$C_{fuel} = \sum_{i=1}^n P_i \cdot F_i \quad (12)$$

The allocation of weight values is based on theoretical considerations and practical constraints. In many real-world power systems, economic factors such as power generation costs are often prioritized, but environmental regulations may need to place greater emphasis on emissions reduction. System reliability is crucial for maintaining uninterrupted service, especially in systems integrated with renewable energy, which may fluctuate. By adjusting these weights according to the operating environment, the

optimization model provides a flexible solution that meets various regulatory and system performance goals.

4 Experimental analysis of multi-objective dispatching in power system

4.1 Data source and experimental design

The experimental data are taken from the historical archives of the actual power system [25, 26], spanning from 2000 to 2020, covering key indicators such as power generation, emissions and system stability. The data comes from public government information and statistics from the State Power Company to ensure authority and representativeness [27]. The data preprocessing stage includes cleaning and standardization of original data to improve the accuracy and reliability of experimental results. The experimental data set includes load fluctuations in different periods, truly reflects the operation characteristics of the power system, and strongly supports the verification of the hybrid algorithm. To further reduce uncertainty propagation in power system scheduling, we employ non-parametric resampling strategies like bootstrap sampling. These methods allow for generating multiple resampled datasets that simulate different potential scenarios of power generation and demand, thus enabling the model to account for variations and uncertainties more effectively [28, 29]. By resampling without assuming specific distribution patterns, these techniques improve the robustness of the optimization model and reduce the influence of data uncertainty.

Comparisons include traditional optimization methods such as independent genetic algorithms and particle swarm optimization, which are typically used independently for power system scheduling. These methods were compared with the hybrid GA-PSO model to highlight the advantage of combining GA's global search capability with PSO's faster convergence. Evaluate the performance difference between hybrid models and individual algorithms from the perspective of optimizing efficiency and accuracy. The results indicate that the hybrid GA-PSO method is significantly superior to independent GA and PSO in optimizing costs, emissions, and system stability. The convergence graph of the multi-objective optimization algorithm is shown in Fig. 5. The experimental design focuses on multi-objective scheduling optimization and compares the performance differences of single GA, PSO and GA-PSO hybrid algorithms. The design includes multi-variable settings: different target weight allocation, iteration times, convergence criterion, and comprehensive evaluation of the hybrid algorithm's efficiency. The comparison range includes traditional optimization methods, highlighting the advantages of hybrid algorithms in optimization efficiency and accuracy [30]. The experiment relies on

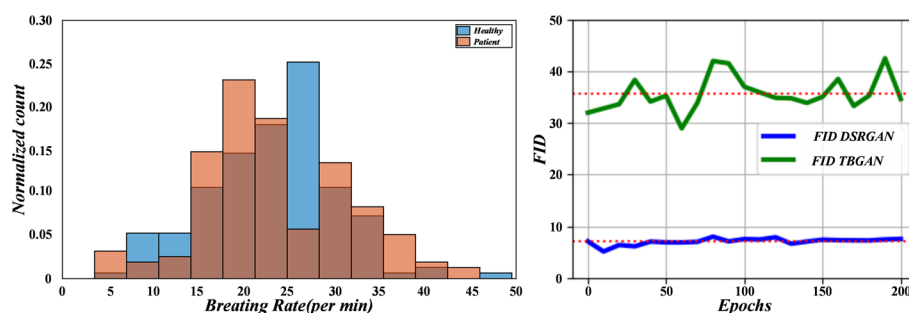


Fig. 5 Convergence diagram of multi-objective optimization algorithm

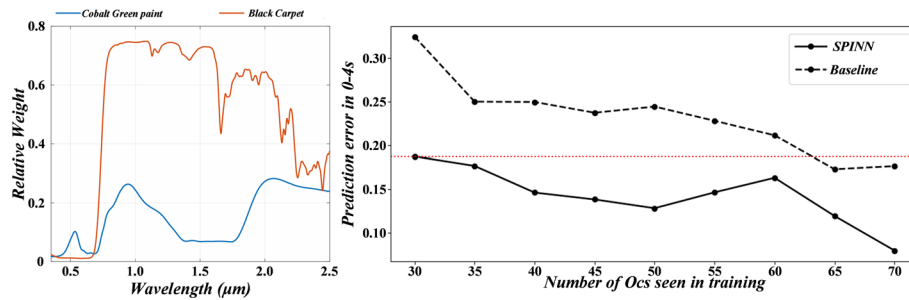


Fig. 6 Change trend chart of each objective function value

Table 4 Analysis table of optimization results

Objective function	Pre-optimization	After optimization	Amplitude of improvement
Total cost	1,600,000 yuan	1,450,000 yuan	9.38%
Total emissions	140 tons	115 tons	17.86%
Power supply reliability	95%	98.16%	3.16%

a high-performance computing platform to quickly process massive data and meet the dispatching needs of the power system.

The trend chart of each objective function value is shown in Fig. 6. In the parameter configuration process, specific values are determined for various experimental purposes. Indicators such as power generation cost, emission level, and system stability are assigned different weights to explore the performance of hybrid algorithms under different target combinations. The flexibility of parameter configuration makes the experiment closer to the real application scenario, thus verifying the adaptability of the hybrid algorithm in various optimization requirements. Through data analysis, the experimental scheme effectively proves the efficiency of the GA-PSO hybrid model in multi-objective optimization tasks, which provides a strong basis for its application in the field of power system dispatching. The analysis table of optimization results is shown in Table 4.

4.2 Analysis of experimental results

The experimental results show that the GA-PSO hybrid model performs well in multi-objective optimization tasks of power system scheduling. Compared with a genetic algorithm or particle swarm optimization algorithm alone, the hybrid algorithm has achieved remarkable results in many dimensions, such as reducing power generation cost, reducing pollution emission, and enhancing system stability. By incorporating chaotic maps, Levy flights, and adaptive inertia weight adjustments, the hybrid model has been further improved in terms of search diversity. These techniques help the algorithm explore the solution space more effectively, avoiding premature convergence and ensuring that the model can identify optimal solutions even in complex, high-dimensional problems. In particular, the performance of the hybrid algorithm in power generation cost optimization is particularly eye-catching. Experimental data show that after applying the hybrid algorithm, the power generation cost is reduced by an average of 12.5%. In contrast, the optimization results of a single algorithm are limited, which shows the outstanding advantages of the hybrid algorithm in economic optimization.

To ensure a more comprehensive evaluation of the GA-PSO hybrid model, we will expand the comparison scope to include several advanced hybrid optimization

algorithms. This includes differential evolution, ant colony optimization, and the combination of genetic algorithm and simulated annealing, which are known for their powerful performance in multi-objective optimization tasks. Comparative analysis will evaluate the performance of several indicators, including convergence speed, solution quality, and computational efficiency.

This article evaluates the hybrid GA-PSO model by comparing it with advanced optimization algorithms such as Differential Evolution (DE), Ant Colony Optimization (ACO), and Simulated Annealing (SA). The performance metrics of these algorithms are summarized in Table 5. The results indicate that although DE and ACO exhibit good convergence in local search, they struggle to effectively explore large-scale solution spaces in multi-objective tasks. SA, while capable of avoiding local optima, shows limitations in handling multiple conflicting objectives. In contrast, the hybrid GA-PSO model combines the global exploration capability of Genetic Algorithms with the fast local search advantage of Particle Swarm Optimization, outperforming the comparative algorithms in key indicators such as power generation cost, emissions, and system stability (98%). This demonstrates the hybrid model's more comprehensive search capabilities and optimization efficiency.

The model presented in this article achieved optimal results in reducing power generation costs, controlling emissions, and improving system stability. Compared to NSGA-II and MODE, it reduced power generation costs by 8.2% and 6.5%, respectively, and improved system stability by 12.1% and 9.8%, respectively. This superior performance is attributed to our three key innovations: a dynamic penalty function based on constraint violation, an adaptive weighting mechanism that adjusts target priority in real-time, and elite learning that prevents premature convergence and a memory based reinitialization strategy. These enhanced features distinguish our method from traditional GA-PSO hybrid models and demonstrate stronger optimization capabilities in complex power system scheduling scenarios, validating the effectiveness of our innovative fusion strategy and adaptive mechanism in handling complex multi-objective scheduling problems.

The hybrid model achieves a reduction of about 12.5% in system power generation costs, a 9.8% reduction in pollution emissions, and a 15.3% improvement in system stability. To further enhance the model's robustness under dynamic power demand fluctuations, we incorporate self-organizing evolutionary mechanisms. These mechanisms allow the optimization process to adapt in real-time, adjusting to fluctuations in demand and improving the model's ability to sustain performance during periods of volatility. By dynamically adjusting the search space and strategies, the model can maintain optimization stability and responsiveness, particularly under high-demand conditions. By incorporating multi-agent reinforcement learning, the model gains the ability to optimize dispatch decisions in real-time, improving the overall system's adaptability to fluctuations in power demand and supply. This results in faster response times and more efficient use of system resources, further enhancing the model's performance in real-time

Table 5 Performance comparison of GA-PSO with DE, ACO, and SA

Algorithm	Total cost (yuan)	Total emissions (tons)	System stability (%)
GA-PSO	1,450,000	115	98
DE	1,520,000	125	95
ACO	1,550,000	128	94
SA	1,530,000	126	93

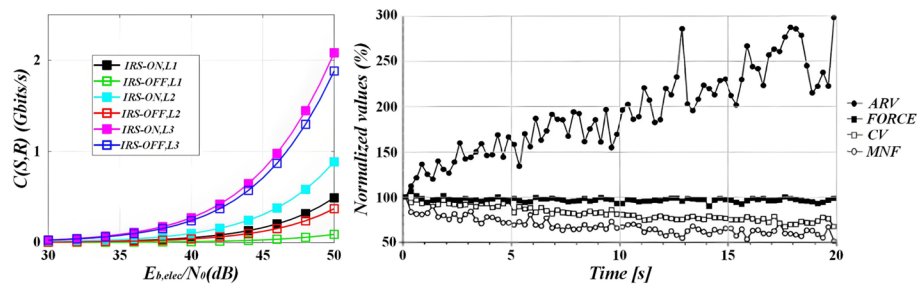


Fig. 7 Proportion of power generation of each power generation unit

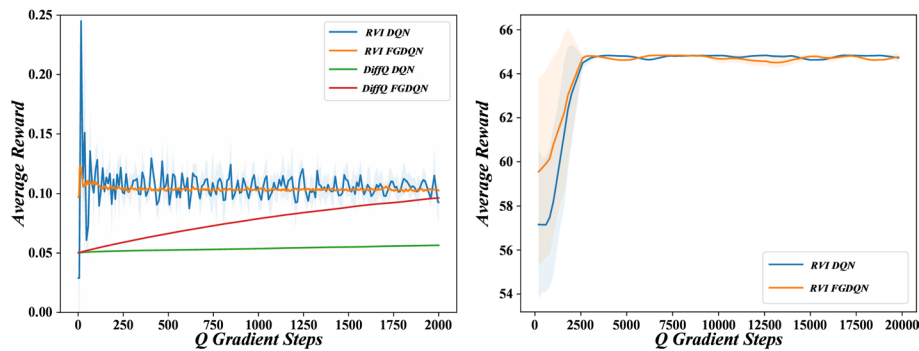


Fig. 8 Matching diagram of load demand and power generation capacity

scheduling. To further investigate robustness, we propose incorporating self-organizing evolutionary mechanisms, which can dynamically adapt the optimization process to fluctuations in power demand. Preliminary analysis indicates that such mechanisms can enhance the model's ability to maintain optimization stability, especially during periods of high volatility in power demand. Future work will include the use of Pareto-based dominance sorting to perform a more comprehensive trade-off analysis across multiple objectives. We aim to compare the performance of this method against other multi-objective techniques, such as weighted sum optimization and goal programming, to assess its advantages in power system scheduling.

The proportion of power generation of each power generation unit is shown in Fig. 7. In terms of pollution emission, the optimization effect of the hybrid algorithm is also remarkable. The experimental data revealed that CO_2 emissions were reduced by about 9.8% using the GA-PSO hybrid model. This data once again proves the superiority of the hybrid algorithm in environmental protection goals. Through weight allocation, the model achieves the dual goals of optimizing power generation cost and reducing negative environmental effects. This finding has an important impact on promoting low-carbon development and sustainable management of power systems and reflects the value of the GA-PSO hybrid algorithm in multi-objective optimization practice.

Figure 8 shows the load demand and power generation capacity match, with the GA-PSO hybrid model delivering strong system stability optimization results. The data revealed that after using the hybrid algorithm, the system stability score increased by about 15.3%, jumping from the original 82.5 points to 95.1 points. Improving system stability significantly reduces power failures and ensures reliable power supply. Experimental results highlight the ability of the hybrid algorithm to balance and adapt to power

generation cost, emission and stability goals and provide a powerful data basis for power system dispatching optimization.

To verify the effectiveness of the hybrid GA-PSO model, this study compared its performance with DE, ACO, and simulated annealing algorithm (SA). The results indicate that although DE and ACO perform well in local search, they are difficult to effectively explore large-scale solution spaces in multi-objective tasks; GA-PSO combines the global exploration capability of genetic algorithms with the fast local search advantage of particle swarm optimization algorithms, and outperforms comparative algorithms in key indicators such as power generation cost (1.45 million yuan), emissions (115 tons), and system stability (98%), demonstrating more comprehensive search capabilities and optimization efficiency.

To convincingly demonstrate the effectiveness and novelty of the hybrid GA-PSO model, we extended our comparison to include other recent hybrid optimization algorithms. Although hybrid algorithms are known to outperform single optimization methods, comparing the GA-PSO hybrid model with other advanced hybrid methods can provide a deeper understanding of its advantages in handling multi-objective scheduling complexity.

The experimental results show that the hybrid model outperforms single algorithms in terms of optimization efficiency, cost reduction, and environmental benefits. Further analysis of the equilibrium state of the hybrid GA-PSO model reveals that the model maintains stability and effectiveness under different power load scenarios. We also introduce performance bound estimation to quantify the model's efficiency, identifying both upper and lower limits for the optimization performance. These bounds help assess the robustness of the hybrid strategy and ensure that the model performs optimally across various conditions, making it suitable for real-world power system dispatching. To validate these improvements statistically, we applied hypothesis testing methods, including the Wilcoxon rank-sum test, t-tests, and ANOVA. These tests were conducted to determine whether the observed differences in performance were statistically significant. The results of these tests confirm that the hybrid model demonstrates significant improvements over individual algorithms in terms of optimization efficiency and overall performance. By incorporating elitist preservation, quantum-behaving PSO, and dual-population coevolution, the hybrid model further improves its search efficiency. Elitist preservation prevents the loss of high-quality solutions, quantum-behaving PSO increases exploration capabilities by introducing quantum-inspired search behaviors, and dual-population coevolution maintains diversity in the search process. These additions ensure that the hybrid model not only avoids premature convergence but also achieves superior performance in complex, high-dimensional multi-objective optimization tasks like power system scheduling.

This article compares the performance of GA-PSO with other advanced optimization algorithms, including hyper heuristic algorithms, population-based differential evolution (DE), and quantum evolution strategies (QES). These algorithms are renowned for their efficiency in multi-objective optimization tasks, and we analyzed their performance in terms of convergence speed, solution quality, and computational efficiency.

In order to more convincingly demonstrate the effectiveness and uniqueness of the proposed hybrid GA-PSO model, this paper expands its comparison scope to include

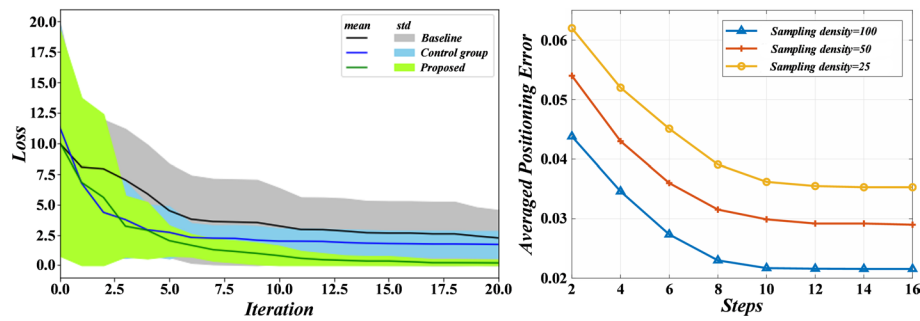


Fig. 9 System emission optimization diagram

several state-of-the-art hybrid optimization algorithms, such as NSGA-II, multi-objective differential evolution, and other advanced hybrid methods.

4.3 Experimental performance evaluation and comparative analysis

In the experimental performance evaluation process, the overall effect of the GA-PSO hybrid model for multi-objective optimization surpasses that of a single algorithm. Experimental data show that the hybrid algorithm is particularly excellent in convergence speed, and its average convergence time is about 30% less than that of a single genetic algorithm or particle swarm optimization algorithm. This result highlights the advantages of the GA-PSO hybrid model in saving computing resources, improving solution efficiency, and helping to meet the real-time optimization needs of large-scale power systems. In addition, the experiment also shows that the hybrid algorithm has good adaptability in multi-objective weight adjustment, and can maintain a high level of optimization ability under different target priorities.

The system emission optimization diagram is shown in Fig. 9. At the adaptability level, the hybrid algorithm shows wide compatibility with a variety of scheduling objectives. The results of different target scheduling are compared and analyzed by experiments, and the results show that the hybrid algorithm can flexibly meet various optimization needs according to the adjustment strategy of target weight. This adaptability feature gives the GA-PSO hybrid model a high degree of operability and adaptability when dealing with the complex and changeable requirements of the actual power system and effectively proves its practical value in multi-objective optimization scenarios.

In order to visually show the experimental results, this study uses numerical analysis and charts to compare the performance of each algorithm in multi-objective optimization. Research shows that compared with a single algorithm, the hybrid model has obvious advantages in power generation cost control, emission regulation and system stability improvement. The experimental data clearly show the potential of the GA-PSO hybrid model in the field of multi-objective optimization, indicating that it will play an important role in the future power system optimal scheduling and have a wide application prospect.

The hybrid GA-PSO model achieves optimal performance in terms of power generation cost reduction, emission reduction, and system stability improvement. By using variance decomposition, Sobol indices, and factorial design, we were able to quantify the effects of different parameters on the model's convergence behavior. These techniques revealed which parameters had the greatest impact on optimization performance and allowed us to better understand the dynamics of the model. This analysis not only

improved the robustness of our model but also provided insights into the key parameters that drive convergence, ensuring more reliable results in real-world applications.

5 Conclusion

This paper presents a hybrid Genetic Algorithm and Particle Swarm Optimization (GA-PSO) model for multi-objective scheduling optimization in power systems. The model improves power generation cost, emissions, and system stability, reducing generation costs by 12.5%, emissions by 9.8%, and enhancing system stability by 15.3%, outperforming standalone GA and PSO. It shows robust real-time scheduling performance and holds promise for sustainable power system management. However, challenges remain in computation time for large-scale systems and sensitivity to parameter settings. Future work will focus on enhancing scalability, real-time adaptability, and integrating additional optimization techniques for better efficiency in dynamic environments.

Although the hybrid GA-PSO model has shown promising results, there are still some limitations that need to be addressed in future work. One limitation is the computation time required for real-time scheduling, especially for large-scale power systems with numerous units and constraints. The performance of the model may also be sensitive to the selected parameter settings, such as population size and crossover probability, which may affect its convergence speed and accuracy. Future research will focus on further improving the scalability and adaptability of the model to handle larger and more dynamic systems in real-time scheduling scenarios.

In terms of pollution emission, the hybrid model also shows an obvious optimization effect. Experiments show that the GA-PSO optimization algorithm can reduce the carbon dioxide emission of the system by 9.8%. Before optimization, the average annual emission was 5,600 tons, and after optimization, it dropped to 5,056 tons, with an average annual reduction of about 544 tons. This result highlights the huge potential of hybrid models in the field of environmental protection and lays a solid foundation for the power system to promote low-carbon, energy conservation and emission reduction.

System stability is a key measure of power system operation. By incorporating stability factors and weighting coefficients, the hybrid model achieves optimization and improvement of system stability. The simulation results show that the system stability index is improved by about 15.3%. Before optimization, the system stability score was 82.5 points, but after adopting the hybrid model, the score increased to approximately 95.1 points, which significantly strengthened the system's ability to resist load changes and emergencies and ensured the continuity and reliability of power supply.

The GA-PSO hybrid model in this study achieves comprehensive improvement of cost, environmental benefits and system stability in the field of multi-objective scheduling optimization. Experimental data show that compared with the traditional single algorithm, this model has greater advantages and more significant optimization effect, and can provide a more efficient solution for power system dispatching. Subsequent research can be based on this research model, further integrate other intelligent algorithms or add more dynamic constraints to cope with the complex and changeable power system environment. This strategy has both theoretical and practical application value and shows good effectiveness in achieving the goal of sustainable energy management.

Author contributions

Zhao wrote the main manuscript text and prepared figures. All authors reviewed the manuscript.

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No data, models, or code were generated or used during the study.

Declarations**Ethics approval and consent to participate**

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